

| SEQ NO                                                                             | REV | NOTE NO. | ITEM NAME                                      | ITEM CODE | SIZE RANGE / SMALL END SIZES                                                                                                                      |     | DESCRIPTION                                                                                                           | REFERENCE / LARGE END SIZES                                                                                            |                                                                               | NOTES      |                                                                                                                                          |     |                |                   |
|------------------------------------------------------------------------------------|-----|----------|------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------|-------------------|
| 10                                                                                 |     |          |                                                |           |                                                                                                                                                   |     | *****VALVES*****                                                                                                      |                                                                                                                        |                                                                               | 812        | NIPPLE TO BE CUT FROM SS SCH XXS PIPE.                                                                                                   |     |                |                   |
| 20                                                                                 |     |          |                                                |           |                                                                                                                                                   |     | ***** MODULAR VALVES *****                                                                                            |                                                                                                                        |                                                                               | 816        | COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.                                                    |     |                |                   |
| 30                                                                                 | U04 |          | MODULAR DBB                                    | K25RC02   | 0.5                                                                                                                                               | 1.5 | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV             | 0                                                                                                                      | 0                                                                             | 817        | USE ONLY FOR TIE IN                                                                                                                      |     |                |                   |
| 40                                                                                 | U04 |          | MODULAR DBB                                    | K25RC08   | 2                                                                                                                                                 | 2   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL GO - FULL SIZE TYPE 2BV             | 0                                                                                                                      | 0                                                                             | B1         | ALL BOLTS ARE METRIC DIA.                                                                                                                |     |                |                   |
| 50                                                                                 | U04 | VAV, 817 | MODULAR DBB                                    | K25CC08   | 3                                                                                                                                                 | 3   | MODULAR VALVE DB&B BALL TYPE 2500# CF LTCS ASTM A350 LF2 CL.1 316SS TRIM METAL GO - FULL SIZE TYPE 2BV                | 0                                                                                                                      | 0                                                                             | B2         | 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10                                                  |     |                |                   |
| 60                                                                                 |     | VAV      | MODULAR DBB                                    | M25HC08   | 2                                                                                                                                                 | 2   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ REDUCING LTCS A350LF2 316SS TRIM METAL GO - FULL SIZE TYPE R2BVR               | 3                                                                                                                      | 3                                                                             | B6         | 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22                                                  |     |                |                   |
| 70                                                                                 | U04 | VAW      | MODULAR DBB                                    | M25RC02   | 0.75                                                                                                                                              | 2   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR      | 0                                                                                                                      | 0                                                                             | BA1        | TO BE USED WITH RAISED FACE FLANGES ONLY.                                                                                                |     |                |                   |
| 80                                                                                 | U04 |          | MODULAR DBB                                    | M25RC08   | 2                                                                                                                                                 | 2   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 316SS TRIM METAL GO - CONTROLLED BORE TYPE 2BVR        | 0                                                                                                                      | 0                                                                             | BA4        | TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.                                                                            |     |                |                   |
| 90                                                                                 | U04 | VAV      | MODULAR DBB                                    | M25HC02   | 2                                                                                                                                                 | 2   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR | 3                                                                                                                      | 3                                                                             | FA8        | USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.                                                                                    |     |                |                   |
| 100                                                                                | U04 | VAW      | MODULAR DBB                                    | M25HC02   | 0.75                                                                                                                                              | 1   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR | 2                                                                                                                      | 2                                                                             | FF5        | FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES                                                                           |     |                |                   |
| 110                                                                                | U04 | VAV, 817 | MODULAR DBB                                    | M25YC02   | 1                                                                                                                                                 | 1   | MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ LTCS A350LF2 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE R2BVR       | 3                                                                                                                      | 3                                                                             | SB6        | USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.                                                      |     |                |                   |
| 120                                                                                | U04 | VAV, 817 | MODULAR DBB                                    | M25YC02L  | 1                                                                                                                                                 | 1   | MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ LTCS A350LF2 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE R2BVR       | 3                                                                                                                      | 3                                                                             | UJ2        | USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.                        |     |                |                   |
| 130                                                                                | U04 | VAV, 817 | MODULAR DBB                                    | K25NC08   | 3                                                                                                                                                 | 3   | MODULAR VALVE DB&B BALL TYPE 2500# CF X RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL GO - FULL SIZE TYPE 2BV        | 0                                                                                                                      | 0                                                                             |            | USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS                                                                  |     |                |                   |
| 140                                                                                | U04 | VAV      | MODULAR DBB                                    | M25HC02L  | 2                                                                                                                                                 | 2   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR | 3                                                                                                                      | 3                                                                             | VA5        | USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.                                                                                        |     |                |                   |
| 150                                                                                | U04 | VAW      | MODULAR DBB                                    | M25HC02L  | 0.75                                                                                                                                              | 1   | MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR | 2                                                                                                                      | 2                                                                             | VAV        | VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.                              |     |                |                   |
| 160                                                                                |     | VB2      |                                                |           |                                                                                                                                                   |     | ***** BALL VALVES *****                                                                                               |                                                                                                                        |                                                                               | VB2        | SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.                                                                     |     |                |                   |
| 170                                                                                | U04 |          | BALL                                           | A25RC07   | 2                                                                                                                                                 | 2   | BALL 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO                                                    | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 180                                                                                | U04 | 817      | BALL                                           | A25CC07   | 3                                                                                                                                                 | 3   | BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO                                                     | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 190                                                                                | U04 | 817      | BALL                                           | A25CC07   | 6                                                                                                                                                 | 6   | BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO                                                     | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 200                                                                                | U04 | 817      | BALL                                           | A25CC07   | 12                                                                                                                                                | 12  | BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO                                                     | 0                                                                                                                      | 0                                                                             | VC1        | SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. |     |                |                   |
| 210                                                                                | U04 |          | BALL                                           | B25RC01   | 0.75                                                                                                                                              | 1.5 | BALL 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM FLOATING SOFT FB LO                                               | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 220                                                                                | U04 | VA5      | BALL                                           | B25RC07   | 2                                                                                                                                                 | 2   | BALL 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO                                                    | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 230                                                                                | U04 | 817      | BALL                                           | B25CC07   | 3                                                                                                                                                 | 3   | BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO                                                     | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 240                                                                                | U04 | 817      | BALL                                           | B25CC07   | 6                                                                                                                                                 | 6   | BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO                                                     | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 250                                                                                | U04 | 817      | BALL                                           | B25CC07   | 12                                                                                                                                                | 12  | BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO                                                     | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 260                                                                                |     |          |                                                |           |                                                                                                                                                   |     | ***** GLOBE VALVES *****                                                                                              |                                                                                                                        |                                                                               |            |                                                                                                                                          |     |                |                   |
| 270                                                                                | U04 |          | GLOBE                                          | L25RC48   | 0.75                                                                                                                                              | 1.5 | GLOBE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL OS&Y HW                                                                    | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 280                                                                                | U04 |          | GLOBE                                          | L25RC41   | 2                                                                                                                                                 | 2   | GLOBE 2500# RTJ LTCS A352 LCC INCONEL OS&Y HW                                                                         | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 290                                                                                | U04 | 817      | GLOBE                                          | L25CC42   | 3                                                                                                                                                 | 3   | GLOBE 2500# CF LTCS A352 LCC INCONEL OS&Y GO                                                                          | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 300                                                                                |     |          |                                                |           |                                                                                                                                                   |     | ***** CHECK VALVES *****                                                                                              |                                                                                                                        |                                                                               |            |                                                                                                                                          |     |                |                   |
| 310                                                                                | U04 | VC1      | CHECK                                          | C25RC50   | 0.75                                                                                                                                              | 1.5 | CHECK PISTON 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM                                                                | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 320                                                                                |     |          | CHECK                                          | C25RC52   | 2                                                                                                                                                 | 2   | CHECK DUAL 2500# RTJ LTCS A352 LCC TRIM 16                                                                            | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 330                                                                                | U04 |          | CHECK                                          | C25RC45   | 2                                                                                                                                                 | 2   | CHECK DUAL WAFER LUG 2500# RTJ LTCS A352 LCC INCONEL TRIM THROU BOLTED                                                | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 340                                                                                | U04 | 817      | CHECK                                          | C25CC52   | 3                                                                                                                                                 | 12  | CHECK DUAL DBL FLG 2500# CF LTCS A352 LCC INCONEL TRIM                                                                | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 350                                                                                |     |          |                                                |           |                                                                                                                                                   |     | *****PIPES*****                                                                                                       |                                                                                                                        |                                                                               |            |                                                                                                                                          |     |                |                   |
| 360                                                                                | U04 | 812      | PIPE                                           | 5358511   | 0.5                                                                                                                                               | 1.5 | PIPE SCH XXS SMLS LTCS A333 GR.6 PE                                                                                   | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 370                                                                                |     |          | PIPE                                           | 5358516   | 2                                                                                                                                                 | 2   | PIPE WT 12.0MM SMLS LTCS A333 GR.6 BE                                                                                 | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 380                                                                                |     |          | PIPE                                           | 5358536   | 3                                                                                                                                                 | 3   | PIPE WT 16.0MM SMLS LTCS A333 GR.6 BE                                                                                 | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 390                                                                                |     |          | PIPE                                           | 5358537   | 4                                                                                                                                                 | 4   | PIPE WT 20.0MM SMLS LTCS A333 GR.6 BE                                                                                 | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 400                                                                                |     |          | PIPE                                           | 5358539   | 6                                                                                                                                                 | 6   | PIPE WT 29.0MM SMLS LTCS A333 GR.6 BE                                                                                 | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 410                                                                                |     |          | PIPE                                           | 5358538   | 12                                                                                                                                                | 12  | PIPE WT 53.0MM SMLS LTCS A333 GR.6 BE                                                                                 | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 420                                                                                |     |          | PIPE                                           | 5358532   | 16                                                                                                                                                | 16  | PIPE WT 66.0MM SMLS LTCS A333 GR.6 BE                                                                                 | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 430                                                                                |     |          |                                                |           |                                                                                                                                                   |     | ***** FITTINGS *****                                                                                                  |                                                                                                                        |                                                                               |            |                                                                                                                                          |     |                |                   |
| 440                                                                                | U04 |          | ELBOW                                          | 5392870   | 0.5                                                                                                                                               | 1.5 | ELL 90 DEG LR XXS LTCS A420 WPL6                                                                                      | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 450                                                                                |     |          | ELBOW                                          | 5392871   | 2                                                                                                                                                 | 2   | ELL 90 DEG LR WT 12.0MM LTCS A420 WPL6                                                                                | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 460                                                                                |     |          | ELBOW                                          | 5392818   | 3                                                                                                                                                 | 3   | ELL 90 DEG LR WT 16.0MM LTCS A420 WPL6                                                                                | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 470                                                                                |     |          | ELBOW                                          | 5392819   | 12                                                                                                                                                | 12  | ELL 90 DEG LR WT 53.0MM LTCS A420 WPL6                                                                                | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
| 480                                                                                |     |          | ELBOW                                          | 5392876   | 0.75                                                                                                                                              | 1.5 | ELL 45 DEG LR XXS LTCS A420 WPL6                                                                                      | 0                                                                                                                      | 0                                                                             |            |                                                                                                                                          |     |                |                   |
|  |     |          | TENGIZCHEVROIL<br>TCO<br>TENGIZ,<br>KAZAKHSTAN |           | CONSTRUCTION                                                                                                                                      |     | CORR.ALLOW.                                                                                                           | SERVICE                                                                                                                | MATERIAL                                                                      | DATE       | BY                                                                                                                                       | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |     |          |                                                |           | ALL SIZES - BUTT WELDED<br>PWHT FOR WT > 19MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE |     | CORR. ALLOW: 1.5 MM                                                                                                   | SOUR SERVICE NACE MR0175 / ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>FLANGE RATING: 2500 RF/CF<br>TEMPERATURE RANGE: -40 / 200°C | 12/16/2024 | BY<br>OZAC                                                                                                                               | U04 | Page 1<br>of 3 | 2500H1            |

| SEQ NO | REV | NOTE NO. | ITEM NAME | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                 | REFERENCE / LARGE END SIZES |     | NOTES |
|--------|-----|----------|-----------|-----------|------------------------------|------|-------------------------------------------------------------|-----------------------------|-----|-------|
| 490    |     |          | ELBOW     | 5392875   | 2                            | 2    | ELL 45 DEG LR WT 12.0MM LTCS A420 WPL6                      | 0                           | 0   |       |
| 500    |     |          | ELBOW     | 5392832   | 3                            | 3    | ELL 45 DEG LR WT 16.0MM LTCS A420 WPL6                      | 0                           | 0   |       |
| 510    | U04 |          | ELBOW     | 5392820   | 6                            | 6    | ELL 90 DEG LR WT 29.0MM LTCS A420 WPL6                      | 0                           | 0   |       |
| 520    |     |          | ELBOW     | 5392833   | 12                           | 12   | ELL 45 DEG LR WT 53.0MM LTCS A420 WPL6                      | 0                           | 0   |       |
| 530    |     |          |           |           |                              |      |                                                             |                             |     |       |
| 530    |     |          | TEE       | 5392879   | 0.75                         | 1.5  | TEE SCH XXS LTCS A420 WPL6                                  | 0                           | 0   |       |
| 540    |     |          | TEE       | 5392889   | 2                            | 2    | TEE WT 12.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 550    |     |          | TEE       | 5392837   | 3                            | 3    | TEE WT 16.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 560    |     |          | TEE       | 5392838   | 12                           | 12   | TEE WT 53.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 570    |     |          | TEE RED   | 5392877   | 0.75                         | 1    | TEE RED SCH XXS LTCS A420 WPL6                              | 1                           | 1.5 |       |
| 580    |     |          | TEE RED   | 5392887   | 0.75                         | 1.5  | TEE RED WT 12.0MM / SCH XXS LTCS A420 WPL6                  | 2                           | 2   |       |
| 590    |     |          | TEE RED   | 5392842   | 1.5                          | 1.5  | TEE RED WT 16.0MM/SCH XXS LTCS A420 WPL6                    | 3                           | 3   |       |
| 600    |     |          | TEE RED   | 5392843   | 2                            | 2    | TEE RED WT 16.0MM/12.0MM LTCS A420 WPL6                     | 3                           | 3   |       |
| 610    |     |          | TEE RED   | 5392844   | 2                            | 2    | TEE RED WT 20.0MM/12.0MM LTCS A420 WPL6                     | 4                           | 4   |       |
| 620    | U04 |          | TEE RED   | 5392845   | 6                            | 6    | TEE RED WT 53.0MM/29.0MM LTCS A420 WPL6                     | 12                          | 12  |       |
| 630    |     |          |           |           |                              |      |                                                             |                             |     |       |
| 630    | U04 |          | CAP       | 5392880   | 0.75                         | 1.5  | CAP BW SCH XXS LTCS A420 WPL6                               | 0                           | 0   |       |
| 640    |     |          | CAP       | 5392883   | 2                            | 2    | CAP WT 12.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 650    |     |          | CAP       | 5392846   | 3                            | 3    | CAP WT 16.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 660    |     |          | CAP       | 5392624   | 6                            | 6    | CAP WT 29.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 670    |     |          | CAP       | 5392847   | 12                           | 12   | CAP WT 53.0MM LTCS A420 WPL6                                | 0                           | 0   |       |
| 680    |     |          |           |           |                              |      |                                                             |                             |     |       |
| 680    | U04 |          | RED CONC  | 5392881   | 0.5                          | 1    | REDUCER CONC SCH XXS LTCS A420 WPL6                         | 1                           | 1.5 |       |
| 690    |     |          | RED CONC  | 5392884   | 0.75                         | 1.5  | REDUCER CONC WT 12.0MM / SCH XXS LTCS A420 WPL6             | 2                           | 2   |       |
| 700    |     |          | RED CONC  | 5392859   | 1.5                          | 1.5  | REDUCER CONC WT 16.0MM/XXS LTCS A420 WPL6                   | 3                           | 3   |       |
| 710    |     |          | RED CONC  | 5392858   | 2                            | 2    | REDUCER CONC WT 16.0MM/12.0MM LTCS A420 WPL6                | 3                           | 3   |       |
| 720    |     |          | RED CONC  | 5392852   | 2                            | 2    | REDUCER CONC WT 20.0MM/12.0MM LTCS A420 WPL6                | 4                           | 4   |       |
| 730    |     |          | RED CONC  | 5392853   | 3                            | 3    | REDUCER CONC WT 20.0MM/16.0MM LTCS A420 WPL6                | 4                           | 4   |       |
| 740    | U04 |          | RED ECC   | 5392882   | 0.5                          | 1    | REDUCER ECC SCH XXS LTCS A420 WPL6                          | 0.75                        | 1.5 |       |
| 750    |     |          | RED ECC   | 5392885   | 0.75                         | 1.5  | REDUCER ECC WT 12.0MM / SCH XXS LTCS A420 WPL6              | 2                           | 2   |       |
| 760    |     |          | RED ECC   | 5392862   | 1.5                          | 1.5  | REDUCER ECC WT 16.0MM/XXS LTCS A420 WPL6                    | 3                           | 3   |       |
| 770    |     |          | RED ECC   | 5392861   | 2                            | 2    | REDUCER ECC WT 16.0MM/12.0MM LTCS A420 WPL6                 | 3                           | 3   |       |
| 780    |     |          | RED ECC   | 5392863   | 1.5                          | 1.5  | REDUCER ECC WT 20.0MM / SCH XXS LTCS A420 WPL6              | 4                           | 4   |       |
| 790    |     |          | RED ECC   | 5392867   | 2                            | 2    | REDUCER ECC WT 20.0MM / 12.0MM LTCS A420 WPL6               | 4                           | 4   |       |
| 800    |     |          | RED ECC   | 5392866   | 3                            | 3    | REDUCER ECC WT 20.0MM / 16.0MM LTCS A420 WPL6               | 4                           | 4   |       |
| 810    |     |          | RED ECC   | 5392864   | 3                            | 3    | REDUCER ECC WT 29.0MM / 16.0MM LTCS A420 WPL6               | 6                           | 6   |       |
| 820    |     |          | RED ECC   | 5392869   | 4                            | 4    | REDUCER ECC WT 37.0MM / 20.0MM LTCS A420 WPL6               | 8                           | 8   |       |
| 830    |     |          | RED ECC   | 5392865   | 6                            | 6    | REDUCER ECC WT 53.0MM / 29MM LTCS A420 WPL6                 | 12                          | 12  |       |
| 840    |     |          | RED ECC   | 5392874   | 8                            | 8    | REDUCER ECC WT 53.0MM / 37MM LTCS A420 WPL6                 | 12                          | 12  |       |
| 850    |     |          | RED ECC   | 5392873   | 8                            | 8    | REDUCER ECC WT 66.0MM / 37.0MM LTCS A420 WPL6               | 16                          | 16  |       |
| 860    |     |          |           |           |                              |      | ***** OLETS *****                                           |                             |     |       |
| 860    |     |          |           |           |                              |      |                                                             |                             |     |       |
| 870    |     |          |           |           |                              |      |                                                             |                             |     |       |
| 870    | U04 |          | OLETS     | 5504285   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1   | 2                           | 2   |       |
| 880    | U04 |          | OLETS     | 5504286   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1   | 3                           | 3   |       |
| 890    | U04 |          | OLETS     | 5504287   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2 CL.1 | 4                           | 4   |       |
| 900    | U04 |          | OLETS     | 5504288   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 53.0MM / SCH XXS LTCS A350 LF2 CL.1 | 12                          | 12  |       |
| 910    | U04 | UJ2      | OLETS     | 5504290   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1   | 2                           | 2   |       |
| 920    | U04 | UJ2      | OLETS     | 5504291   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1   | 3                           | 3   |       |
| 930    | U04 | UJ2      | OLETS     | 5504292   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2 CL.1 | 4                           | 4   |       |
| 940    | U04 | UJ2      | OLETS     | 5504293   | 0.75                         | 1.5  | NIPOFLANGE 2500# RTJ WT 53.0MM / SCH XXS LTCS A350 LF2 CL.1 | 12                          | 12  |       |
| 950    | U04 |          | OLETS     | 5504294   | 1                            | 1    | NIPOFLANGE 2500# RTJ WT 29.0MM / SCH XXS LTCS A350 LF2 CL.1 | 6                           | 6   |       |
| 960    | U04 | UJ2      | OLETS     | 5504294   | 1.5                          | 1.5  | NIPOFLANGE 2500# RTJ WT 29.0MM / SCH XXS LTCS A350 LF2 CL.1 | 6                           | 6   |       |
| 970    | U04 |          | OLETS     | 5504289   | 1                            | 1    | NIPOFLANGE 2500# RTJ WT 66.0MM / SCH XXS LTCS A350 LF2 CL.1 | 16                          | 16  |       |
| 980    | U04 | FF5      | OLETS     | 5504522   | 1                            | 1    | NIPOFLANGE FLAT BOTTOM 2500# RTJ SCH XXS LTCS A350 LF2 CL.1 | 0                           | 0   |       |
| 990    |     |          |           |           |                              |      |                                                             |                             |     |       |
| 990    | U04 |          | OLETS     | 5505158   | 0.75                         | 1    | WELDOLET WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1               | 2                           | 2   |       |
| 1000   | U04 |          | OLETS     | 5505159   | 0.75                         | 1    | WELDOLET WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1               | 3                           | 3   |       |
| 1010   | U04 |          | OLETS     | 5505160   | 0.75                         | 1    | WELDOLET WT 20.0 MM / SCH XXS LTCS A350 LF2 CL.1            | 4                           | 4   |       |
| 1020   | U04 |          | OLETS     | 5505156   | 2                            | 2    | WELDOLET WT 29.0MM/WT 12.0MM LTCS A350 LF2 CL.1             | 6                           | 6   |       |
| 1030   | U04 |          | OLETS     | 5505161   | 0.75                         | 1    | WELDOLET WT 53.0 MM / SCH XXS LTCS A350 LF2 CL.1            | 12                          | 12  |       |
| 1040   | U04 |          | OLETS     | 5505162   | 2                            | 2    | WELDOLET WT 53.0 MM / 12.0MM LTCS A350 LF2 CL.1             | 12                          | 12  |       |
| 1050   | U04 |          | OLETS     | 5505163   | 3                            | 3    | WELDOLET WT 53.0 MM / 16.0MM LTCS A350 LF2 CL.1             | 12                          | 12  |       |
| 1060   | U04 |          | OLETS     | 5504377   | 0.75                         | 0.75 | WELDOLET FLAT BOTTOM SCH XXS LTCS A350 LF2 CL.1             | 0                           | 0   |       |
| 1070   |     |          |           |           |                              |      |                                                             |                             |     |       |
| 1070   | U04 | UJ2      | ELBOFLG   | 5516025   | 1.5                          | 1.5  | ELBOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1   | 2                           | 2   |       |
| 1080   | U04 | UJ2      | ELBOFLG   | 5516026   | 1.5                          | 1.5  | ELBOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1   | 3                           | 3   |       |
| 1090   | U04 | UJ2      | ELBOFLG   | 5516027   | 1.5                          | 1.5  | ELBOFLANGE 2500# RTJ WT 20.0MM/SCH XXS LTCS A350 LF2 CL.1   | 4                           | 4   |       |
| 1100   | U04 | UJ2      | ELBOFLG   | 5516028   | 1.5                          | 1.5  | ELBOFLANGE 2500# RTJ WT 53.0MM / SCH XXS LTCS A350 LF2 CL.1 | 12                          | 12  |       |
| 1110   |     |          |           |           |                              |      | ***** FLANGES *****                                         |                             |     |       |
| 1110   |     | 816      |           |           |                              |      |                                                             |                             |     |       |
| 1120   |     |          |           |           |                              |      |                                                             |                             |     |       |
| 1120   | U04 |          | FLANGE    | 5557706   | 0.5                          | 1.5  | FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH XXS                 | 0                           | 0   |       |

|  | TENGIZCHEVROI<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                                                      | CORR.ALLOW.         | SERVICE                                                                                                                | MATERIAL                                                                      | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|                                                                                    |                                               | ALL SIZES - BUTT WELDED<br>PWHT FOR WT > 19MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW: 1.5 MM | SOUR SERVICE NACE MR0175 / ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>FLANGE RATING: 2500 RF/CF<br>TEMPERATURE RANGE: -40 / 200°C | 12/16/2024 | BY<br>OZAC | U04 | Page 2<br>of 3 | 2500H1            |

| SEQ NO | REV | NOTE NO. | ITEM NAME  | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                                 | REFERENCE / LARGE END SIZES |     | NOTES |
|--------|-----|----------|------------|-----------|------------------------------|------|-----------------------------------------------------------------------------|-----------------------------|-----|-------|
| 1130   | U04 |          | FLANGE     | 5619890   | 2                            | 2    | FLG WN 2500# RTJ LTCS A350 LF2 CL.1 WT 12.0MM                               | 0                           | 0   |       |
| 1140   | U04 |          | FLANGE     | 5619936   | 3                            | 3    | FLG WN 2500# RTJ LTCS A350 LF2 CL.1 WT 16.0MM                               | 0                           | 0   |       |
| 1150   | U04 | 817      | FLANGE     | 5619893   | 3                            | 3    | FLG COMPACT 2500# RJ LTCS A350 LF2 CL.1 WT 16.0MM                           | 0                           | 0   |       |
| 1160   | U04 |          | FLANGE     | 5557718   | 6                            | 6    | FLG WN 2500# RF LTCS A350 LF2 CL.1 WT 29.0MM                                | 0                           | 0   |       |
| 1170   | U04 | 817      | FLANGE     | 5619894   | 12                           | 12   | FLG COMPACT 2500# RJ LTCS A350 LF2 CL.1 WT 53.0MM                           | 0                           | 0   |       |
| 1180   | U04 |          | FLANGE     | 5557714   | 3                            | 3    | FLG WN 2500# RF LTCS A350 LF2 CL.1 WT 16.0MM                                | 0                           | 0   |       |
| 1190   | U04 |          | FLANGE     | 5557715   | 4                            | 4    | FLG WN 2500# RF LTCS A350 LF2 CL.1 WT 20.0MM                                | 0                           | 0   |       |
| 1200   | U04 |          | FLANGE     | 5557716   | 12                           | 12   | FLG WN 2500# RF LTCS A350 LF2 CL.1 WT 53.0MM                                | 0                           | 0   |       |
| 1210   | U04 | FA8      | TEE RED    | 5599713   | 0.5                          | 0.75 | FLG SCRDR (FNPT) REDUCING 2500# RTJ LTCS A350 LF2 CL.1                      | 0.75                        | 1.5 |       |
| 1220   |     |          | FLANGE     | 5851202   | 0.75                         | 0.75 | FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE                                | 0                           | 0   |       |
| 1230   |     |          | FLANGE     | 5851202   | 2                            | 2    | FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE                                | 0                           | 0   |       |
| 1240   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1240   | U04 |          | BLIND      | 5615615   | 0.75                         | 2    | FLG BLIND 2500# RTJ LTCS A350 LF2 CL.1                                      | 0                           | 0   |       |
| 1250   | U04 |          | BLIND      | 5615617   | 3                            | 12   | FLG BLIND 2500# RF LTCS A350 LF2 CL.1                                       | 0                           | 0   |       |
| 1260   | U04 | 817      | BLIND      | 5619795   | 3                            | 3    | FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 CL.1 LINE CLASS 2500H1             | 0                           | 0   |       |
| 1270   | U04 | 817      | BLIND      | 5619795   | 6                            | 6    | FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 CL.1 LINE CLASS 2500H1             | 0                           | 0   |       |
| 1280   | U04 | 817      | BLIND      | 5619795   | 12                           | 12   | FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 CL.1 LINE CLASS 2500H1             | 0                           | 0   |       |
| 1290   | U04 | 817      | FLANGE     | 5563349   | 6                            | 6    | FLG COMPACT 2500# RJ LTCS A350 LF2 CL.1 WT 29.0MM LINE CLASS 2500H1         | 0                           | 0   |       |
| 1300   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1300   |     |          |            |           |                              |      | ***** PIPE NIPPLES *****                                                    |                             |     |       |
| 1310   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1310   |     | 812      |            | 0         |                              |      | NIPPLE BBE CUT FROM PIPE                                                    |                             |     |       |
| 1320   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1320   |     |          |            |           |                              |      | ***** GASKETS *****                                                         |                             |     |       |
| 1330   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1330   |     |          | GASKET     | 5672271   | 0.5                          | 12   | GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR                                 | 0                           | 0   |       |
| 1340   |     |          | GASKET     | 5674480   | 0.5                          | 3    | GASKET 2500# RTJ SS OCTAGONAL RING                                          | 0                           | 0   |       |
| 1350   | U04 | 817      | GASKET     | 5673520   | 3                            | 12   | GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 2500H1 | 0                           | 0   |       |
| 1360   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1360   | U04 |          | GASKET     | SP-2990   | 6                            | 6    | FLANGE GASKET HIGH INTEGRITY DESGN FIRE SAFE ASME 2500 RAISED FACE FLANGES  | 0                           | 0   |       |
| 1370   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1370   |     | B1       |            |           |                              |      | ***** BOLTING *****                                                         |                             |     |       |
| 1380   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1380   | U04 |          | BOLTING    | 5675770   | 0.5                          | 8    | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS                           | 0                           | 0   |       |
| 1390   | U04 | BA4      | BOLTING    | 5675770   | 10                           | 12   | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS                           | 0                           | 0   |       |
| 1400   |     | BA1      | BOLTING    | 5675776   | 10                           | 99   | STUD-BOLT WITH 2 NUTS API-6A718                                             | 0                           | 0   |       |
| 1410   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1410   |     | SB6      | BOLTING    | 5675675   | 12                           | 12   | JACK SCREW HEX HEAD A320 GR L7M                                             | 0                           | 0   |       |
| 1420   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1420   |     |          |            |           |                              |      | ***** BLINDS & SPACERS *****                                                |                             |     |       |
| 1430   |     |          |            |           |                              |      |                                                                             |                             |     |       |
| 1430   |     |          | SPEC BLIND | 5749590   | 0.75                         | 2    | SPECTACLE BLIND 2500# RTJ LTCS A516-GR70                                    | 0                           | 0   |       |
| 1440   |     |          | SPEC BLIND | 5749597   | 3                            | 3    | SPECTACLE BLIND 2500# RF LTCS A516-GR70                                     | 0                           | 0   |       |
| 1450   | U04 | 817      | SPEC BLIND | 5749598   | 3                            | 3    | SPECTACLE BLIND COMPACT 2500# RJ LTCS A516-GR70                             | 0                           | 0   |       |
| 1460   |     |          |            |           |                              |      |                                                                             |                             |     |       |

|                                                                                    |                                               |                                                                                                                                                   |                     |                                                                                                                        |                                                                               |            |            |     |                |                   |
|------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|  | TENGIZCHEVROI<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                                                      | CORR.ALLOW.         | SERVICE                                                                                                                | MATERIAL                                                                      | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |                                               | ALL SIZES - BUTT WELDED<br>PWHT FOR WT > 19MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW: 1.5 MM | SOUR SERVICE NACE MR0175 / ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>FLANGE RATING: 2500 RF/CF<br>TEMPERATURE RANGE: -40 / 200°C | 12/16/2024 | BY<br>OZAC | U04 | Page 3<br>of 3 | 2500H1            |